

Project Overview

NewsToday is a web-based news aggregation platform that allows users to view news articles across multiple categories such as Technology, Business, Science, Health, Entertainment, and more. The application emphasizes personalized user experience through preferences and provides functionality to manage user profiles.

The main objective of this project was to **demonstrate API integration and Flask backend skills**.

Technologies Used

Layer	Technology	Purpose
Backend	Python 3	Core programming language
Backend Framework	Flask	Web framework to manage routes, templates, and sessions
Templates	Jinja2	Dynamic HTML rendering with data injection
Styling	Bootstrap 5	Responsive UI styling
API	NewsAPI	Fetch latest news articles dynamically
Data Storage	JSON (data.json)	Stores user credentials, preferences
Security	werkzeug.security	Password hashing for secure storage
File Handling	Flask request.files,	Upload and store profile pictures
	secure_filename	
Frontend Features	HTML/CSS/JS layout	Form validation, buttons, and interactive

Key Features

- 1. **User Authentication** ○ Signup and Login system with **hashed passwords** for secure storage.
 - Session management with Flask’s session.
- 2. **User Profile** ○ Display of user information: ID, Name, Email.
 - Ability to edit name, email.

3. **Personalized News** ○ Users select preferences for news categories. ○
Home page displays news based on selected preferences.
 - Each news article links to the source article in a new tab.
4. **News Navigation** ○ A dedicated "NewsToday" page displays all categories.
 - Users can filter news dynamically by category.
5. **Search Functionality**
 - **Search bar** allows users to directly search for a news category of choice.
6. **Responsive UI** ○ Bootstrap ensures mobile-friendly and responsive layout.
 - Cards and badges used for clean visualization.
7. **Flash Messages**
 - Success and error messages for actions like login, signup, and profile updates.
8. **Secure Password Handling**
 - Passwords are stored **hashed using werkzeug.security**, preventing plain-text storage.

Limitations

1. **Data Storage**
 - Project uses data.json instead of MySQL or other databases to keep things simple and easy to understand for learning purposes ○ Uses data.json instead of a proper database. may not scale well.
 - No real-time database updates or concurrent handling for multiple users.
 2. **Security** ○ No email verification or password recovery features.
 - Session security is basic (Flask's default session handling).
 3. **Limited input validation**
 - The project has minimal or no input validation, which could allow incorrect or unsafe data to be entered.
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Pros

1. Fully functional **Flask backend** with routes for login, signup, profile, news.
 2. **Dynamic API integration** for fetching live news.
 3. Personalized experience through **user preferences**.
 4. Interactive UI with **Bootstrap cards, badges, and forms**.
 5. **Profile management** including.
 6. **Flash messages** improve user feedback for actions.
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Cons

1. Using JSON file for storage limits scalability and reliability.
 2. Minimal frontend customization beyond default Bootstrap components.
 3. Profile image handling is rudimentary.
 4. No advanced security features (e.g., email verification, password reset).
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Future Improvements

1. **Database Integration**
 - Replace data.json with SQLite/PostgreSQL for better scalability.
 2. **Enhanced Security** ○ Add password reset, email verification, and two-factor authentication.
 - Implement HTTPS for secure communication.
 3. **Advanced Features**
 - Search bar to filter articles. ○ Save favorite articles for users.
 - Paginate news articles to improve performance.
 4. **Performance**
 - Cache API results to reduce API calls and improve load time.
 - Optimize image handling and compression.
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Conclusion

NewsToday successfully demonstrates **integration of APIs with a Flask backend**, **dynamic content rendering with Jinja2**, and **basic user management**. It provides a solid foundation for learning web development, API usage, session management, and responsive frontend design. With enhancements such as a proper database, improved security, and richer UI, it can evolve into a fully-featured news aggregation platform.

Images







